

Introduction to Machine Learning

Machine learning is a subset of artificial intelligence that enables systems to learn and improve from experience without being explicitly programmed. It focuses on developing programs that can access data and use it to learn for themselves.

Supervised Learning

Supervised learning is the most common type of machine learning. The algorithm is trained on labelled data, meaning each training example is paired with an output label. Common algorithms include linear regression, decision trees, and support vector machines.

Neural Networks Overview

Neural networks are computing systems inspired by biological neural networks in the brain. They consist of layers of interconnected nodes. Deep learning uses neural networks with many layers to learn complex representations of data.

Model Evaluation

Model evaluation is the process of assessing how well a machine learning model performs. Common metrics include accuracy, precision, recall, and F1 score for classification tasks, and mean squared error for regression tasks.